Exhaustive Audit of the Tax Calculator Implementation and Cross-Border Assumptions

1. Bottom-Line Verdict and Executive Summary

The current implementation of the automated tax-calculator repository represents a highly sophisticated programmatic framework for managing complex, cross-border cryptocurrency transactions and domestic self-employment tax obligations. By executing a modular pipeline that ingests raw exchange ledgers, normalizes them, and processes them through distinct Polish tax engines (PIT-38, PIT-36, and PIT-28), the repository automates a vast amount of intricate fiscal mathematics. However, an adversarial, forensic review of its underlying legal assumptions, data processing pipelines, and form outputs reveals a severe dichotomy between its computational prowess and its legal defensibility.

The implementation correctly executes the mechanical aspects of Polish tax law. It accurately applies the Polish annual cost-pooling mechanism for capital gains under PIT-38, systematically categorizes trading commissions versus non-deductible exchange fees, and properly models the standard mathematical deductions for the Polish flat-rate tax (ryczałt) applicable to IT services.

Conversely, the repository harbors critical structural and theoretical fragilities that expose the taxpayer to extreme audit risks. The most severe legal vulnerability is the repository's foundational assumption regarding the cross-border portability of barter-generated cost basis. Specifically, the inclusion of pre-residency, salary-paid USD Coin (USDC) into the Polish PIT-38 cost pool is an aggressive, legally unsupported extrapolation of Polish domestic law applied to a foreign jurisdiction. While the taxpayer possesses strong factual evidence of these receipts, the statutory mechanism to convert foreign-taxed income into domestic Polish capital costs is absent.

Furthermore, the implementation suffers from significant data custody issues. The reliance on manual text proxies for critical exchange data (such as Coinbase), combined with structural vulnerabilities in handling Binance CSV exports, compromises the forensic integrity of the cost-basis chain. Finally, the PIT-36 and PIT-28 reporting engines contain procedural and statutory omissions, notably the failure to attach the mandatory PIT/ZG annex for foreign-source income and the misapplication of invoice issue dates for revenue recognition under ryczalt rules. The repository requires immediate architectural partitioning of high-risk cost pools, integration of raw source data, and correction of statutory timing rules before the generated artifacts can be safely submitted to the Polish National Revenue Administration (KAS).

2. Comprehensive Data Audit

The integrity of any automated tax calculation is entirely dependent on the completeness, immutability, and structural validity of the ingested data. The current data architecture of the repository mixes primary-source ledger exports with manually compiled proxies, creating a fractured chain of custody that will not withstand scrutiny during a formal tax audit (Czynności Sprawdzające or Postępowanie Podatkowe).

2.1. Primary-Source Data Inventory and Validity

The repository successfully utilizes primary source data for a portion of its operations, which forms the reliable core of the normalized dataset (outputs/normalized_all_exchanges.csv, totaling 1,938 rows).

The utilization of the Kraken yearly ledger CSVs (1,087 rows) is highly robust. Kraken's ledger format provides a complete chronological accounting of trades, fees, and transfers, maintaining internal reference IDs that allow the normalization engine to accurately pair trade legs and isolate exact fee structures. Similarly, the utilization of the FTX file (core_transactions.csv yielding 97 rows) represents a defensible best-effort primary source given the exchange's subsequent insolvency and operational collapse. Under the circumstances of exchange failure, Polish tax authorities generally accept contemporary ledger exports if they distinctly parse fiat deposits and trading fees, which this file appears to do.

The Binance raw XLSX exports (754 normalized rows) present a more complex scenario. While they are primary source documents, Binance's CSV generation logic—specifically the "Generate All Statements" function—is notorious within the forensic accounting community for separating trades into micro-fills and frequently omitting or misattributing margin and futures realized Profit and Loss (PnL).[1] If the normalization script src/tax_calc/normalizers/binance.py relies on grouping these micro-fills without referencing a definitive, aggregated order ID, it risks creating phantom taxable events or miscalculating the basis.[4]

2.2. Missing Data and Proxy Vulnerabilities

The most glaring deficiency in the data pipeline is the total absence of a raw Coinbase CSV export. The repository currently relies on coinbase-purchases.txt and celsius-simplex-purchases.txt, which are manually structured text files. The parser (src/tax_calc/normalizers/salary.py) processes these manual files using a simple repeating block format containing a URL, date, amount, asset, and optional fee lines.

Relying on hand-entered proxies for live PIT-38 cost inputs is procedurally unacceptable in an adversarial tax audit. Polish tax law strictly dictates the evidentiary requirements for documenting capital costs.[5] KAS routinely demands unedited, original exchange logs, trade confirmation PDFs, and corresponding bank statements to verify the exact timing and fiat outflow of cryptocurrency acquisitions.[6] Manual files strip away essential cryptographic metadata, internal transaction routing IDs, and exact timestamps (down to the second) necessary to defend the application of the correct National Bank of Poland (NBP) exchange

rate from the preceding business day.[8] A tax inspector will view a manual .txt file as an arbitrary assertion rather than documentary evidence, leading to the summary disallowance of those costs.

Similarly, the 2022-fantom-salary.txt and 2022-2023-pre-poland-salary.txt files act as manual bridges for on-chain income. While the taxpayer possesses underlying on-chain evidence, smart contract interactions, and Swedish tax returns to support these receipts [9], the translation of this high-fidelity evidence into a manual text file introduces a human intermediary into the automated pipeline, breaking the programmatic chain of custody.

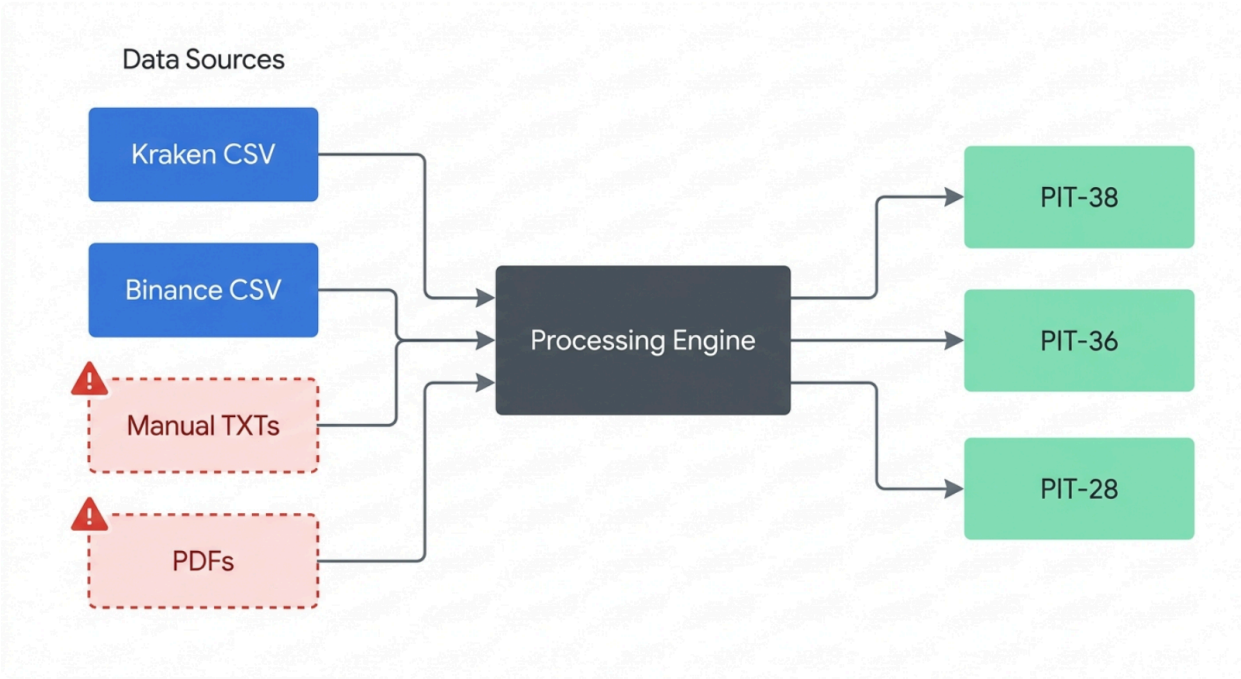
2.3. Unparsed Invoices and Business Income Documents

The docs/invoices/ directory reveals a partially completed pipeline for business income. The 2025/invoices-clearstar/ directory contains 7 XML invoices that the code successfully parses for the PIT-28 calculation. However, the 2025/invoices-conclave/ directory contains 7 PDF invoices that are currently ignored by the code.

If these Conclave invoices represent JDG (Jednoosobowa Działalność Gospodarcza) revenue, the PIT-28 calculation is currently under-reporting gross receipts, which carries severe penal fiscal consequences. If they represent pre-JDG personal income, the PIT-36 calculation is incomplete. All revenue-generating artifacts must be machine-readable and ingested into the core pipeline to ensure the outputs reflect the true, holistic economic reality of the 2025 tax year.

Furthermore, the repository houses 8 ZUS DRA PDFs for 2025-05 through 2025-12. These documents are correctly utilized to feed the PIT-28 health-insurance deduction assumptions, as Polish law permits the deduction of 50% of paid health contributions for taxpayers on the ryczalt scheme.[10]

Data Pipeline Integrity and Audit Risk Areas



The current data pipeline relies on verifiable ledger exports for Kraken and Binance, but introduces severe audit vulnerabilities through manual text proxies for Coinbase purchases and historical salary records. These manual nodes break the chronological chain of custody required by Polish tax authorities.

3. Legal Audit of Core Assumptions

The core of this external audit rests on evaluating the repository's programmatic translation of Polish tax law, particularly regarding the complex intersection of cross-border residency transitions and cryptocurrency cost basis pooling. The implementation must strictly adhere to the Polish Personal Income Tax (PIT) Act and the prevailing interpretive doctrine of the National Revenue Information (KIS).

3.1. Pre-Residency Fiat Purchases on Exchanges

Verdict: Correct.

The repository correctly assumes that documented fiat-to-crypto acquisition costs incurred while the taxpayer was a tax resident of Sweden can be carried forward into the Polish PIT-38 cost pool, provided those specific assets were not disposed of (consumed) prior to the move to Poland.

Polish tax law, specifically Art. 22 ust. 14 of the PIT Act, establishes that costs of obtaining revenues from the odpłatne zbycie (onerous disposal) of virtual currencies are documented expenses incurred directly for their acquisition.[5] The law does not mandate that the taxpayer

must have been a Polish tax resident at the exact moment the fiat was spent. Recent individual tax rulings (e.g., 0112-KDIL2-1.4011.796.2025.2.TR) explicitly confirm this position: costs of acquiring cryptocurrencies incurred abroad can be recognized as tax-deductible costs after a change of residency to Poland, provided they are properly demonstrated and the subsequent disposal of those assets occurs while the individual is subject to unlimited Polish tax liability.[6]

The calculator's logic to parse the celsius-simplex-purchases.txt and ftx-purchases.txt (representing actual fiat outlays) and absorb them into the 2023 PIT-38 carry-forward pool is theoretically sound, though, as noted in the data audit, the physical evidence standard must be upgraded from .txt proxies to raw bank/exchange statements.

3.2. Pre-Residency Salary-Paid USDC

Verdict: Incorrect / Extremely High Risk.

The repository currently aggregates pre-residency salary paid in USDC (derived from 2022-fantom-salary.txt and 2022-2023-pre-poland-salary.txt) into the Polish PIT-38 cost pool. This is the single riskiest assumption in the codebase and represents a highly aggressive, legally unsustainable position that must be classified as an critical vulnerability.

To understand why this fails, one must dissect the Polish barter doctrine. Under Polish law, when a Polish tax resident receives cryptocurrency as remuneration for services, it is treated as a barter transaction.[13] The PLN equivalent of the cryptocurrency received is taxed as income at the moment of receipt (e.g., under Art. 13 pkt 8 for personal services).[14] Simultaneously, this legal fiction establishes an acquisition cost for the cryptocurrency equal to that exact same PLN value, which the taxpayer can then utilize in PIT-38 upon future disposal to prevent double taxation.[13]

However, the repository attempts to stretch this Polish domestic barter doctrine backward in time, applying it to income earned and taxed in Sweden, while the taxpayer had absolutely no tax nexus to Poland. Polish authorities strictly require proof of an *expense incurred* ("wydatki poniesione") for the direct acquisition of the virtual currency.[5] In a barter scenario, the "expense" is the provision of the service. Because the service was provided, the value was established, and the income was realized entirely outside Polish tax jurisdiction, the KAS will argue that no qualifying "expense" was incurred under the purview of Polish law that can be carried forward as a capital cost.

While the taxpayer possesses excellent evidentiary backing—including on-chain transfers, employment contracts, and Swedish tax returns proving the asset was taxed [9]—this evidence merely proves the *factual* reality of the receipt. It does not fulfill the *statutory* requirement of Art. 22 ust. 14 under Polish jurisdiction. There is no general step-up in basis upon immigrating to Poland. The Polish tax authority permits the importation of actual out-of-pocket capital expenditures (fiat spent to buy crypto) [6], but they are highly unlikely to recognize an "intangible" cost basis generated through a foreign statutory mechanism (Swedish taxation of employment income) without an explicit, personalized protective ruling from KIS.

If this bucket is disqualified during an audit, the taxpayer will face a total denial of that cost

basis, resulting in a flat 19% tax on the entire gross proceeds of the USDC disposal, plus severe default interest penalties. This data must be excised from the pre-2023 cost pool.

3.3. Post-Residency Salary-Paid USDC (Jan-Apr 2025)

Verdict: Correct.

For USDC salary received *after* Polish residency was established on 2023-04-12, but *before* the commencement of the JDG in 2025-05, the repository correctly models the dual nature of the transaction. Stablecoins are definitively treated as virtual currencies under Art. 5a pkt 33a of the PIT Act.[14]

1. **Income Recognition:** The receipt of stablecoins for prop-trading or IT services outside a registered business framework is correctly classified under Art. 13 pkt 8 of the PIT Act (działalność wykonywana osobiście - personally performed activity).[14] The application of the 20% statutory cost deduction (Koszty Uzyskania Przychodu - KUP) is standard and correct for this specific revenue source, as it represents remuneration for services utilizing personal skill.
2. **Cost Basis Generation:** The repository correctly assumes that the gross PLN value recognized as income immediately establishes an equivalent acquisition cost for the PIT-38 pool under the established barter doctrine.[13] There is no double-tax or double-deduction issue here; this is the precise, intended statutory mechanism to prevent taxing the same economic value twice.[14] The income is taxed on the tax scale (PIT-36), and the subsequent disposal of the crypto is taxed at 19% (PIT-38) only on any capital appreciation above the receipt value.

3.4. Stablecoin Deposits as PIT-38 Costs

Verdict: Incorrect / Double-Counting Risk.

The repository currently contains logic (a salary_years dedup mechanism) designed to treat stablecoin deposits to exchanges as PIT-38 cost events if they are not covered by explicitly mapped salary data for that year. This is a dangerous and legally flawed heuristic.

Under Polish law, transferring assets between wallets or depositing them onto an exchange does not constitute a taxable event, nor does it establish a cost basis.[17] The cost basis is established at the exact, chronological moment the stablecoin is *acquired*—either purchased with fiat currency or received as legally recognizable salary.

Treating a deposit as a proxy for an acquisition cost inherently inflates the cost pool if that same asset was already accounted for elsewhere, or if it was acquired at a different historical price point. The Polish tax authority strictly requires evidence of the actual acquisition transaction (the transfer of fiat or the rendering of a service), not merely the funding of a centralized exchange account.[5] This heuristic must be deprecated immediately in favor of strict, chronological matching of actual acquisition events.

3.5. Fee Policy (Trade, Funding, and Withdrawal)

Verdict: Correct.

The repository's fee parsing logic accurately reflects current Polish jurisprudence and the strict delineations of Art. 22 ust. 14:

- **Trade Fees:** Commissions and exchange fees paid directly on taxable buy/sell rows are correctly deducted as direct costs of acquisition or sale.[19] If you buy 1 BTC for 100,000 PLN and pay a 100 PLN fee, your cost basis is 100,100 PLN.
- **Funding Fees:** The programmatic exclusion of funding fees (margin fees, perpetual swap funding rates, loan interest) is absolutely correct. Art. 22 ust. 14 explicitly prohibits deducting costs related to financing the purchase of virtual currencies.[5] While a recent Supreme Administrative Court (NSA) ruling allowed certain indirect costs for real estate to be spread over time [21], this broader commercial doctrine does not override the explicit, highly specific statutory exclusion for crypto financing costs codified in the PIT Act.
- **Withdrawal Fees:** The exclusion of withdrawal and network transfer fees is correct. Moving assets between one's own wallets or exchanges is not a taxable disposal. Consequently, the network or exchange fees incurred to facilitate these internal transfers are generally viewed by KAS as non-deductible personal or administrative expenses, not direct costs of asset acquisition.[18]

Deductibility of Cryptocurrency Fees under Polish PIT-38

Fee Type	Definition	Deductible	Statutory Basis	Calculator Status
Trading Fees	Commissions on Buy/Sell	YES	Direct Cost of Acquisition	Included
Funding & Margin Fees	Costs to finance leverage	NO	Art 22 ust 14 Exclusion	Excluded
Withdrawal Fees	Network fees for wallet transfers	NO	Not a disposal event	Excluded

Under Polish tax law (Art. 22 ust. 14 of the PIT Act), only direct costs of acquisition and sale are deductible. The repository correctly includes exchange trading commissions while properly excluding network transfer fees and financing costs (such as margin or funding fees).

Data sources: Podatki.gov.pl, Infor.pl, Divly.com

3.6. Valuation Fallback Logic

Verdict: Probably Correct but Risky.

The repository prefers NBP rates when a fiat or stablecoin counterparty exists and falls back to CoinGecko historical PLN prices for pure crypto-to-crypto internal tracking.

According to Art. 11a ust. 2 of the PIT Act, revenue and expenses incurred in foreign currencies must be converted into Polish Zloty (PLN) using the average exchange rate announced by the NBP from the last business day preceding the day the revenue was obtained or the cost incurred.[8]

For pure crypto-to-crypto swaps, there is no taxable event in Poland.[8] Therefore, the exact PLN valuation at the moment of a swap is largely irrelevant for tax purposes, making the CoinGecko fallback harmless for internal portfolio tracking. However, for any taxable event—such as a crypto-to-fiat trade, or the receipt of USDC salary as barter payment—the strict NBP mechanism must be utilized.

Using CoinGecko as a fallback for missing fiat pairs during a taxable event is an acceptable emergency accounting mechanism, but in a formal audit, KAS will demand a statutory cross-rate calculation. The implementation should ensure that whenever a direct fiat pair is

missing during a taxable event, it queries the USD or EUR value of the asset on the preceding day, and then manually applies the official *preceding day's NBP USD/PLN or EUR/PLN rate*, rather than relying on a direct, third-party CoinGecko PLN pair calculation which may deviate from the statutory NBP table.

4. Implementation Audit

The Python codebase, primarily originating from src/tax_calc/cli.py and branching into the various normalizers, was reviewed for logical coherence and strict adherence to the stated tax principles.

4.1. Exchange Normalization Heuristics

The normalization engine's ability to accurately translate chaotic exchange outputs into unified transaction rows is paramount. The current normalized dataset yields 1,938 rows.

Transaction Type	Row Count	Legal Treatment in PIT-38 Engine
staking_reward	346	Excluded (Taxable only upon sale to fiat) [20]
deposit	274	Excluded (Neutral event) [18]
swap_out / swap_in	216 / 216	Excluded (Crypto-to-crypto is non-taxable) [8]
withdrawal	177	Excluded (Neutral event) [18]
buy	166	Included (Establishes Cost Basis) [5]
sell	128	Included (Establishes Taxable Revenue) [5]
fiat_withdrawal	120	Excluded (Neutral event) [18]
internal_transfer	88	Excluded (Neutral event) [18]

The exclusion of transaction classes like internal_transfer, earn_allocation, fiat_deposit, fiat_withdrawal, funding_fee, and unknown from the PIT-38 calculation is **fully correct** under Polish law. Fiat deposits and withdrawals are purely banking events; taxation only occurs upon

the explicit exchange of crypto for fiat, goods, or services.[22]

However, the specific normalizers have distinct fragilities:

- **Binance (src/tax_calc/normalizers/binance.py):** The grouped trade reconstruction is computationally dangerous. Binance APIs and CSVs often decouple fees from the core trade or split a single market order into dozens of micro-fills at varying prices.[1] If the code arbitrarily groups entries that combine unrelated legs, it will average out the cost basis improperly. Furthermore, if futures/margin trading occurred, the code must rely exclusively on the explicit Realized P&L and Margin Fees data points, rather than attempting to reconstruct leveraged positions from raw buy/sell quantities, which will fail mathematically.[2]
- **Kraken (src/tax_calc/normalizers/kraken.py):** The Kraken ledger reconstruction is generally robust. Kraken's inherent refid system allows the Python logic to cleanly pair trade legs and accurately attribute the fee asset to the correct transaction.
- **Coinbase (src/tax_calc/normalizers/coinbase.py):** Currently non-functional as it relies entirely on the manual fallback text file, which has been identified as a critical audit risk.

4.2. The PIT-38 Engine and Annual Cost Pooling

The cost_pool.py and pit38.py modules accurately capture the essence of the Polish cryptocurrency tax regime. Unlike jurisdictions like the United States or Sweden that require First-In-First-Out (FIFO) or Specific Identification lot tracking, Poland employs a simplified, aggregate annual cost pooling system for virtual currencies.[5]

Under this system, all documented acquisition costs incurred in a calendar year are aggregated (reported in Kolumna c of PIT-38). If these total costs exceed the total revenue generated from crypto-to-fiat disposals in that same year, the resulting loss is not written off; rather, the excess cost is carried forward indefinitely to the subsequent tax year (reported in Kolumna d).[5]

The engine correctly identifies that crypto-to-crypto swaps are non-taxable [8] and seamlessly models the arithmetic carry-forward from 2022 into the active Polish tax years of 2023, 2024, and 2025. The code logically accumulates allowable buy costs and deducts allowable fees.

However, the mathematical perfection of the engine is entirely undermined by the input parameters. Because the engine processes the high-risk, pre-residency USDC salary bucket, the initial seed value of the carry-forward pool is artificially massive. This foundational error infects the entire subsequent mathematical chain, creating a false sense of tax security.

4.3. The PIT-36 Engine

The pit36.py module accurately models the general scale tax parameters for the Jan-Apr 2025 USDC payments categorized as personal income before the establishment of the JDG. It correctly applies the statutory 20% KUP deduction to the gross income [14] and integrates the standard tax-free allowance (kwota wolna od podatku, currently 30,000 PLN).

The module executes the core mathematics flawlessly. However, its scope is overly narrow. It

computes the domestic liability but fails to incorporate the international reporting elements required for income originating from foreign entities, a critical oversight detailed further in the Form-Output audit.

4.4. The PIT-28 Engine

The pit28.py module is designed to calculate the JDG ryczalt income based on the 2025 Clearstar invoices.

- **Rate Selection:** The application of the 12% flat rate for IT and software services is standard, accurate, and safe based on current KIS interpretations for programmers.[25]
- **Deduction Treatment:** The engine correctly deducts 50% of the paid health insurance contributions from the revenue base prior to calculating the tax.[10]
- **Revenue Date Rule:** The implementation contains a critical error in its timing rule. The code currently sets the revenue date effectively as the invoice issue date, converting the EUR values using the NBP EUR/PLN rate from the last business day before that issue date. This contradicts Art. 14 ust. 1c of the PIT Act. For continuous IT services settled in defined periods (e.g., monthly retainers), revenue arises on the last day of the settlement period (usually the last day of the month), regardless of whether the invoice is issued a few days later.[27] Using the issue date instead of the service completion date applies the wrong historical exchange rate, misstating the PLN revenue.

5. Form-Output Audit

The ultimate test of the repository is whether the generated artifacts (pit38_results.json, markdown reports) map correctly to the official Polish tax forms designed by the Ministry of Finance.

5.1. PIT-38 Outputs

The mapping of the repository's data to the PIT-38 fields is computationally accurate based on the internal variables, perfectly tracking the sequential logic of the form:

- **Poz. 34 (Przychody):** Total taxable revenue from crypto-to-fiat disposals.
- **Poz. 35 (Koszty poniesione w roku podatkowym):** Direct acquisition costs incurred in the current calendar year.
- **Poz. 36 (Koszty z lat ubiegłych):** The carry-forward line. The engine correctly shifts the previous year's Poz. 38 (Nieodliczone koszty) directly into this field.[5]
- **Poz. 37 (Dochód):** Correctly calculated as Poz. 34 - (Poz. 35 + Poz. 36). If the result is negative, it reports zero, as crypto capital losses cannot offset other income types.
- **Poz. 38 (Koszty do przeniesienia):** Correctly calculated as (Poz. 35 + Poz. 36) - Poz. 34.

Tax Year	Poz 34 (Revenue)	Poz 35 (New Costs)	Poz 36 (Prior Costs)	Poz 37 (Income)	Poz 38 (Carry-Forwar

					d)
2023	411,143.91	703,072.73	668,757.92	0.00	960,686.74
2024	785,508.57	488,065.61	960,686.74	0.00	663,243.77
2025	*Not provided*	271,316.95	663,243.77	197,371.25	589,298.07

The arithmetic mapping is perfect. For example, in 2024, the taxpayer realized massive revenue (785,508.57 PLN). The costs incurred that year (488,065.61) plus the massive carry-forward from 2023 (960,686.74) completely shielded the revenue, resulting in zero taxable income and leaving 663,243.77 PLN to carry forward to 2025.

However, this exposes the catastrophic risk detailed in Section 3.2. The entire defensive wall protecting the massive 2024 revenues is built upon the 668,757.92 PLN carry-forward entering 2023. Because this number is artificially inflated by pre-residency Swedish USDC salary, KAS will likely disallow it. If KAS strikes out 500,000 PLN of that initial carry-forward, the taxpayer will suddenly face a massive, unshielded tax liability in 2024, complete with penal interest for underpayment.

5.2. PIT-36 Outputs

The pit36_report_2025.md contains the correct core mathematics for Art. 13 pkt 8 income:

- Gross income: 162,735.75 PLN
- 20% costs: 32,547.15 PLN
- Taxable income: 130,188.60 PLN
- Tax base after kwota wolna: 100,189 PLN
- Tax due: 14,060 PLN

However, the output is fatally flawed from a filing compliance perspective because it entirely omits the **PIT/ZG annex**.[30] Since the USDC payments originated from a foreign entity (or a decentralized protocol with a foreign nexus), the taxpayer is legally obligated to report the foreign-source income on the PIT/ZG attachment, mapped to the specific country of the payer.[30]

Furthermore, the repository must account for the specific double-taxation avoidance method dictated by the relevant Double Taxation Treaty (DTT) and the Multilateral Instrument (MLI).

DTT Interaction (Sample)	Method Applicable in 2025	Impact on PIT-36
Sweden (if payer domicile)	Exemption with progression	Poland taxes only if no Swedish tax paid, but affects Polish rate.
Cyprus (if payer domicile)	Exemption with progression	[32]
China (if payer domicile)	Proportional deduction	[32]

For prop-trading or freelance income physically performed while the taxpayer is sitting in Poland, Poland usually retains primary taxation rights regardless of the payer's location, meaning full tax is due in Poland. The PIT/ZG annex is still required to declare the origin of the funds, ensuring the return is not rejected as formally incomplete.

5.3. PIT-28 Outputs

The pit28_report_2025.md correctly maps total revenue (267,794.30 PLN), calculates the 12% tax before deductions (32,135 PLN), applies the 50% health deduction (2,923.84 PLN derived from the 5,847.67 PLN total) [10], and outputs the final tax due (29,211 PLN). The output form structure is procedurally sound. However, as noted in the implementation audit, the underlying revenue figure of 267,794.30 PLN will slightly shift once the NBP date-fetching logic is corrected to reflect service completion dates rather than invoice issue dates.

6. Corrections and Remediation Directives

To elevate the repository from an experimental computational framework to a legally defensible tax compliance tool, the following immediate corrections must be implemented in the codebase.

6.1. High-Priority: Quarantine Pre-Residency USDC

- **The Rule:** The repository must immediately cease automatically dumping pre-residency USDC salary into the 2023 PIT-38 cost pool. The Polish barter doctrine cannot retroactively apply to foreign-taxed income to create a domestic capital cost basis.
- **Code Path Change:** Modify src/tax_calc/normalizers/salary.py and src/tax_calc/cost_pool.py. Implement logic to evaluate the date of the salary receipt against the taxpayer's established Polish residency date (2023-04-12).
    - If date < 2023-04-12, tag the row with a new attribute: foreign_jurisdiction_barter.
    - Explicitly exclude foreign_jurisdiction_barter from Poz. 35 and Poz. 36 inputs.
- **Estimated Impact:** This will systematically gut the 668,757.92 PLN carry-forward entering 2023. Consequently, the tax liability in 2024 will swing from 0.00 PLN to a significant,

actual tax due, as the massive 785,508.57 PLN revenue will no longer be shielded by phantom costs.

6.2. High-Priority: Enforce Strict Statutory Revenue Dates for PIT-28

- **The Rule:** Ryczalt revenue from continuous IT services must be converted using the NBP rate from the business day preceding the *last day of the settlement period* (service execution), not the date the invoice was generated.[27]
- **Code Path Change:** Modify src/tax_calc/pit28.py. When parsing the XML and PDF invoices, extract both issue_date and service_period_end_date. Use min(issue_date, service_period_end_date, payment_receipt_date) as the anchor date. Fetch the NBP rate for anchor_date - 1 business day.
- **Estimated Impact:** Shifts the PLN valuation of the monthly EUR invoices. This may result in slightly higher or lower final tax depending on FX macro fluctuations, but it guarantees statutory compliance and prevents KAS from rejecting the books.

6.3. Medium-Priority: Implement PIT/ZG Generation

- **The Rule:** Foreign-source income reported on PIT-36 requires a PIT/ZG annex detailing the jurisdiction of origin.[30]
- **Code Path Change:** Extend src/tax_calc/pit36.py to output a secondary JSON block mapping the foreign income to a specific ISO country code (e.g., USA, SWE, MLT) based on the payer's corporate domicile. Update outputs/pit36_report_2025.md to include a designated section explicitly outputting the required fields for the PIT/ZG form.
- **Estimated Impact:** Purely procedural compliance. Prevents KAS from initiating a time-consuming administrative summons (Wezwanie) to correct a formally defective return.

6.4. Medium-Priority: Deprecate Stablecoin Deposit Cost Proxy

- **The Rule:** Exchange deposits are not taxable events and cannot be used as proxies for acquisition costs.[18]
- **Code Path Change:** Remove the salary_years dedup logic entirely. Require explicit matching to a fiat purchase ledger or a valid, post-residency salary receipt ledger.
- **Estimated Impact:** Lower carry-forward, higher forensic accuracy. Eliminates the risk of double-counting costs if a salary payment was temporarily held in a self-custody cold wallet before being sent to an exchange months later.

6.5. Data Substitution: Coinbase CSV

- **The Rule:** Replace manual text proxies with verifiable exchange ledgers to maintain an unbroken cryptographic and chronological chain of custody.[7]
- **Action:** The user must manually export the raw Coinbase statement CSV from the exchange platform. Implement src/tax_calc/normalizers/coinbase.py to parse this specific

CSV schema, extracting Buy, Sell, and Fee columns. Delete coinbase-purchases.txt.

7. Risk Ranking and Filing Strategy

If the taxpayer proceeds with filing the generated outputs in their current state, they invite severe regulatory exposure and penal fiscal consequences. The issues are ranked below by order of financial severity and likelihood of audit detection by KAS algorithms.

1. **Extreme Risk - Pre-Residency USDC Salary in PIT-38:**
    - *Status:* Must Be Corrected Before Filing.
    - *Reasoning:* The Polish tax authority systematically cross-references high PIT-38 cost carry-forwards (Poz. 36) against declared historical income on PIT-36/37. If a taxpayer claims over 660,000 PLN in costs imported from 2022, but has no Polish tax history establishing that barter income in 2022, an automated red flag is generated. The reliance on a Swedish tax nexus to establish a Polish domestic cost basis is an untested, highly aggressive legal theory that will fail upon inspection, resulting in massive back-taxes and statutory interest.
2. **High Risk - Manual Proxy Data for Coinbase:**
    - *Status:* Must Be Corrected Before Filing.
    - *Reasoning:* In the event of a routine desk audit (Czynności Sprawdzające), the taxpayer has 7 to 14 days to provide the underlying documentation for Poz. 35 (costs incurred). Submitting a manually typed text file will be immediately rejected as invalid accounting evidence, leading to a summary denial of those costs and an upward revision of taxable income.[7]
3. **Medium-High Risk - PIT-28 Revenue Recognition Dates:**
    - *Status:* Correct Before Next Filing.
    - *Reasoning:* Using the invoice issue date rather than the service completion date [27] alters the NBP exchange rate applied. While the net tax difference may be negligible, a systemic, automated error in applying statutory timing rules indicates flawed accounting practices and can trigger a deeper, more hostile audit of the JDG's books.
4. **Medium Risk - Omission of PIT/ZG:**
    - *Status:* Correct Before Filing.
    - *Reasoning:* This is a procedural error. The financial calculations for PIT-36 are sound, but the tax return will be deemed formally incomplete without the annex detailing the foreign-source nature of the USDC prop-trading income.[30]
5. **Medium Risk - Binance Grouped Trades:**
    - *Status:* Requires Code Audit/Testing.
    - *Reasoning:* Binance's CSV exports frequently merge unrelated trades or truncate leveraged liquidations.[4] If the normalizer script fails to catch a truncation, the taxpayer may accidentally under-report revenue or over-report costs.
6. **Medium Risk - Stablecoin Deposit Proxy:**
    - *Status:* Correct Before Next Filing.

- *Reasoning:* Using deposits as proxies for costs creates a false inflation of the cost pool that will crumble if KAS demands proof of the actual acquisition expenditure.[5]

In conclusion, the tax-calculator repository is a powerful framework that accurately translates complex Polish tax mathematics into code. However, its baseline legal assumption regarding the cross-border portability of barter-generated cost basis is deeply flawed. The taxpayer must recognize that while out-of-pocket fiat expenditures made in Sweden can safely migrate to the Polish tax regime, non-fiat compensation earned and taxed under Swedish jurisdiction cannot be seamlessly retrofitted into Polish PIT-38 as deductible capital costs. By executing the recommended data integrations and restricting the cost-pool logic strictly to post-residency events and verifiable fiat outlays, the repository will transition from a high-risk liability into a robust, audit-ready compliance engine.

Works cited

1. Binance Taxes: What's Reported, What's Taxable, and How People Overpay - Awaken Tax, accessed on April 16, 2026, https://awaken.tax/integrations/binance-taxes
2. Binance futures : r/CryptoTax - Reddit, accessed on April 16, 2026, https://www.reddit.com/r/CryptoTax/comments/1oo33pw/binance_futures/
3. Adding your Binance account using a CSV file - Recap Help Center, accessed on April 16, 2026, https://help.recap.io/en/articles/10485229-adding-your-binance-account-using-a-csv-file
4. Binance Transaction History CSV Import (Grouping by day issue) - Koinly Support, accessed on April 16, 2026, https://discuss.koinly.io/t/binance-transaction-history-csv-import-grouping-by-day-issue/1413
5. Zbycie kryptowalut - podatki.gov.pl, accessed on April 16, 2026, https://www.podatki.gov.pl/podatki-osobiste/pit/informacje-podstawowe/co-jest-opodatkowane/zbycie-kryptowalut/
6. Interpretacja indywidualna z dnia 2 grudnia 2025 r., Dyrektor ..., accessed on April 16, 2026, https://www.inforlex.pl/dok/tresc,FOB0000000000007475065,Interpretacja-indywidualna-z-dnia-2-grudnia-2025-r-Dyrektor-Krajowej-Informacji-Skarbowej-sygn-0112-KDIL2-1-4011-796-2025-2-TR.html
7. Skutki podatkowe obrotu kryptowalutami w PIT, VAT i PCC - Aktualności - Urząd Skarbowy w Kościerzynie - Izba Administracji Skarbowej w Gdańsku, accessed on April 16, 2026, https://www.pomorskie.kas.gov.pl/urzad-skarbowy-w-koscierzynie/wiadomosci/aktualnosci/-/asset_publisher/z8KF/content/id/7798510
8. Podatek od kryptowalut 2026: jak rozliczyć PIT-38?, accessed on April 16, 2026, https://kancelariamw.pl/jak-rozliczyc-podatek-od-kryptowalut/
9. Podatek dochodowy od osób pracujących w Szwecji i zamieszkałych za granicą Polska Zasady podatkowe dla osób zamieszkałych - Skatteverket, accessed on

April 16, 2026, https://www.skatteverket.se/download/18.35c34f651660af3747c31b/1537881977102/skv_442_pl.pdf
10. Odliczenie składki zdrowotnej w PIT - e-pity, accessed on April 16, 2026, https://www.e-pity.pl/ulgi-odliczenia/odliczenie-skladki-zdrowotnej-pit/
11. Odliczenie składek na ubezpieczenie zdrowotne PIT - podatki.gov.pl, accessed on April 16, 2026, https://www.podatki.gov.pl/ulgi-i-odliczenia/odliczenie-skladek-na-ubezpieczenie-zdrowotne-pit/
12. Wydatki na komputerową kopalnię bitcoinów nie są kosztem podatkowym - Prawo.pl, accessed on April 16, 2026, https://www.prawo.pl/podatki/kopalnia-kryptowalut-a-koszty-podatkowe,517054.html
13. 0113-KDIPT2-3.4011.34.2022.2.SJ, Koszty uzyskania przychodu ze sprzedaży walut wirtualnych - Pismo wydane przez: Dyrektor... - ustawy, accessed on April 16, 2026, https://sip.lex.pl/orzeczenia-i-pisma-urzedowe/pisma-urzedowe/0113-kdipt2-3-4011-34-2022-2-sj-koszty-uzyskania-185196009
14. 0115-KDIT1.4011.745.2025.3.MN, Skutki podatkowe wynagrodzenia w formie stablecoinów, accessed on April 16, 2026, https://sip.lex.pl/orzeczenia-i-pisma-urzedowe/pisma-urzedowe/0115-kdit1-4011-745-2025-3-mn-skutki-podatkowe-185306952
15. Skutki podatkowe obrotu kryptowalutami w PIT, VAT i PCC - Komunikaty - Drugi Urząd Skarbowy w Gdyni - Izba Administracji Skarbowej w Gdańsku, accessed on April 16, 2026, https://www.pomorskie.kas.gov.pl/drugi-urzad-skarbowy-w-gdyni/wiadomosci/komunikaty/-/asset_publisher/y0lU/content/id/7798510
16. [Przychody z działalności wykonywanej osobiście] - Art. 13. - Podatek dochodowy od osób fizycznych. - Dz.U.2025.163 t.j. - ustawy, accessed on April 16, 2026, https://sip.lex.pl/akty-prawne/dzu-dziennik-ustaw/podatek-dochodowy-od-osob-fizycznych-16794311/art-13
17. Kryptowaluta jako środek płatniczy? Skutki podatkowe zapłaty kryptowalutą za towary lub usługi - Accreo, accessed on April 16, 2026, https://accreo.pl/kryptowaluta-jako-srodek-platniczy-skutki-podatkowe-zaplaty-kryptowaluta-za-towary-lub-uslugi/
18. Nie wypłaciłeś środków z giełdy – czy musisz rozliczyć kryptowaluty? - Divly, accessed on April 16, 2026, https://divly.com/pl/przewodniki/czy-wyplata-z-gieldy-ma-znaczenie-podatki-krypto
19. Podatki od kryptowalut - Infor, accessed on April 16, 2026, https://g.infor.pl/p/_files/37110000/1-podatki-od-kryptowalut-37110199.pdf
20. Rozliczenia PIT od kryptowalut – obowiązki podatkowe inwestorów - MDDP, accessed on April 16, 2026, https://www.mddp.pl/rozliczenia-pit-od-kryptowalut-i-obowiazki-podatkowe-z-t

ym-zwiazane/
21. Precedensowy wyrok NSA w sprawie rozliczania pośrednich kosztów uzyskania przychodów - MDDP, accessed on April 16, 2026, https://www.mddp.pl/precedensowy-wyrok-nsa-w-sprawie-rozliczania-posrednich-kosztow-uzyskania-przychodow/
22. Podatek od kryptowalut - jak rozliczyć PIT od krypto w 2025/2026 - e-pity, accessed on April 16, 2026, https://www.e-pity.pl/podatek-pit-od-kryptowalut/
23. Binance API vs CSV difference in transactions - Koinly Support, accessed on April 16, 2026, https://discuss.koinly.io/t/binance-api-vs-csv-difference-in-transactions/11796
24. Binance nightmare... How are you using Koinly to tackle Binance tx? - #41 by jon_cina, accessed on April 16, 2026, https://discuss.koinly.io/t/binance-nightmare-how-are-you-using-koinly-to-tackle-binance-tx/3647/41
25. Ryczałt, a może IP BOX dla programisty lub informatyka? - PITax, accessed on April 16, 2026, https://www.pitax.pl/wiedza/poradnik-rozliczenia/ryczalt-a-moze-ip-box-dla-programisty-lub-informatyka/
26. Ryczałt dla programisty – jakie stawki ryczałtu mogą dotyczyć branży IT? | Blog, accessed on April 16, 2026, https://solidnaksiegowa.com/ryczalt-dla-programisty-jakie-stawki-ryczaltu-moga-dotyczyc-branzy-it/
27. 0111-KDIB1-3.4010.677.2025.1.PC, Moment powstania przychodu z usług rozliczanych w okresach rozliczeniowych - ustawy, accessed on April 16, 2026, https://sip.lex.pl/orzeczenia-i-pisma-urzedowe/pisma-urzedowe/0111-kdib1-3-4010-677-2025-1-pc-moment-powstania-185308839
28. [Przychody z pozarolniczej działalności gospodarczej] - Art. 14. - Podatek dochodowy od osób fizycznych. - Dz.U.2025.163 t.j. - ustawy, accessed on April 16, 2026, https://sip.lex.pl/akty-prawne/dzu-dziennik-ustaw/podatek-dochodowy-od-osob-fizycznych-16794311/art-14
29. Art. 14. 1. Za przychód z działalności, o której mowa w art. 10 ust. 1 pkt 3, uważa się kwoty należne, choćby nie zostały faktycznie otrzymane, po wyłączeniu wartości zwróconych towarów, udzielonych bonifikat i skont. U podatników dokonujących sprzedaży towarów i usług opodatkowanych podatkiem od towarów i usług za przychód z tej, accessed on April 16, 2026, https://przepisy.gofin.pl/przepisyno,700,178875,0,0,20180124,2,0.html
30. PIT/ZG: Jak rozliczyć dochody zagraniczne w zeznaniu PIT? - Szlachetna Paczka, accessed on April 16, 2026, https://www.szlachetnapaczka.pl/jedenprocent/artykuly/dochody-zagraniczne-w-pit-36-zg/
31. PIT ZG - instrukcja krok po kroku, jak wypełnić, dla kog... - PITax, accessed on April 16, 2026, https://www.pitax.pl/wiedza/poradnik-rozliczenia/pit-zg-zalacznik-z-informacja-o-dochodach-z-zagranicy/

32. Wykaz metod unikania podwójnego opodatkowania - Wskaźniki i Stawki - GOFIN.pl, accessed on April 16, 2026, https://wskazniki.gofin.pl/wskaznik/453/wykaz-metod-unikania-podwojnego-opodatkowania
33. Informacja w sprawie podstawy wymiaru składki oraz kwoty składki na ubezpieczenie zdrowotne w 2025 r. dla niektórych grup ubezpieczonych - ZUS, accessed on April 16, 2026, https://www.zus.pl/-/informacja-w-sprawie-podstawy-wymiaru-sk%C5%82adki-oraz-kwoty-sk%C5%82adki-na-ubezpieczenie-zdrowotne-w-2025-r.-dla-niekt%C3%B3rych-grup-ubezpieczonych